Data Analyst Interview Challenge 2026

Findings Report

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Databricks Challenge: Data Profiling samples.nyctaxi.trips, results and anomalies

Check	Result
Total number of rows	21,932
First and Last pickup	2016-01-01 and 2016-02-29
trip_distance = 0	76
fare_amount <= 0	10
Dropoffs done before the pickup	1
Maximum fare amount	275
Longest trip (in hours)	~23.98

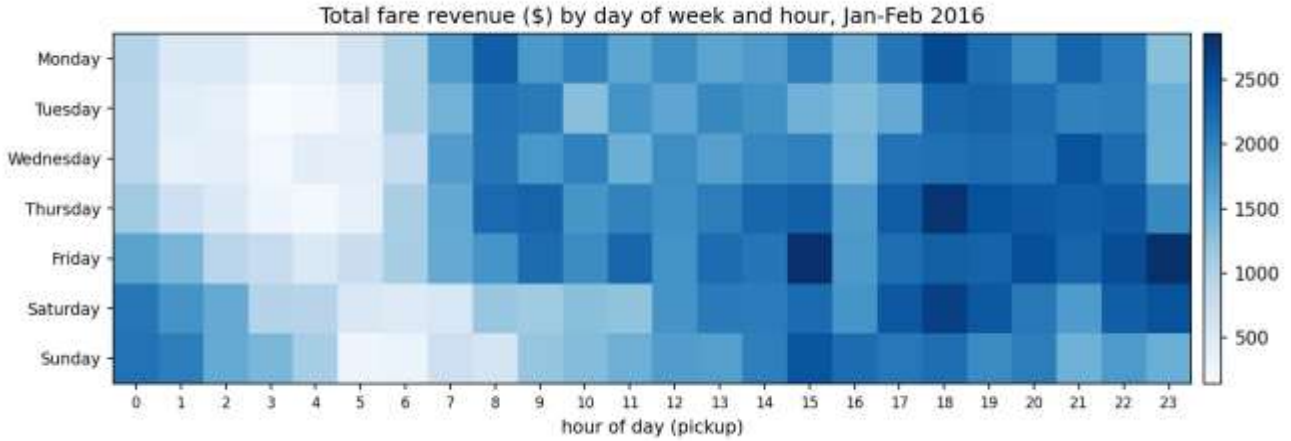


Figure 1: Taxi Revenue Heatmap by Day and Hour

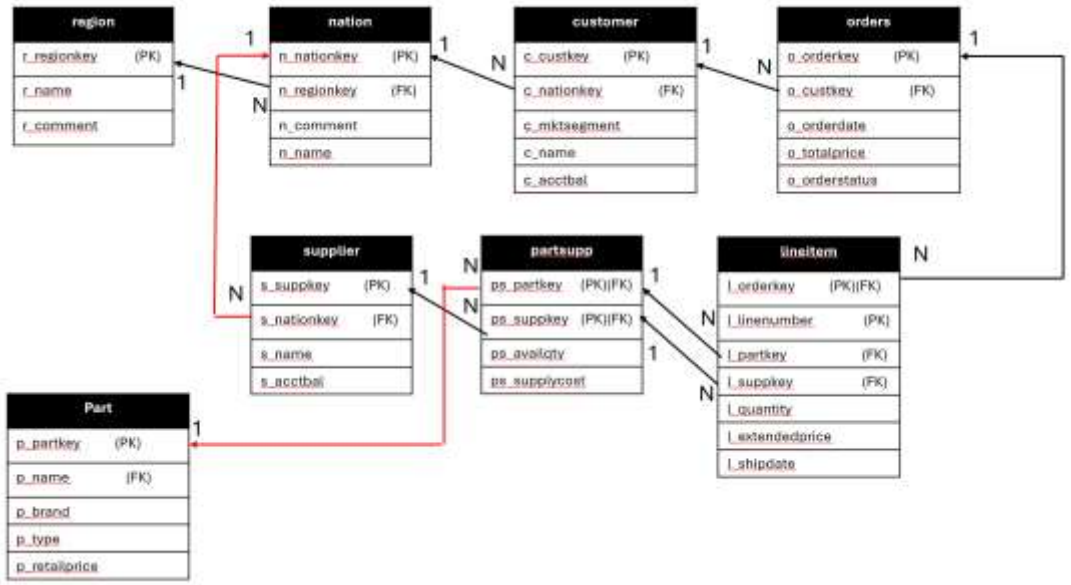
Anomalies found:

- 10 zero or negative fares
- 76 zero-distance trips, 19 of them with fares over $50
- 31 trips longer than 12 hours (16 are almost exactly 24 hours, probably a wrong dropoff time)

Revenue:

- Total revenue $270,832; best slot Friday 15:00 ($2,856)
- Friday is the top day ($44,743) and Tuesday the lowest ($34,248); evenings from about 18:00 to 20:00 bring the most

Databricks Challenge: Entity relationship diagram for the tables samples.tpch



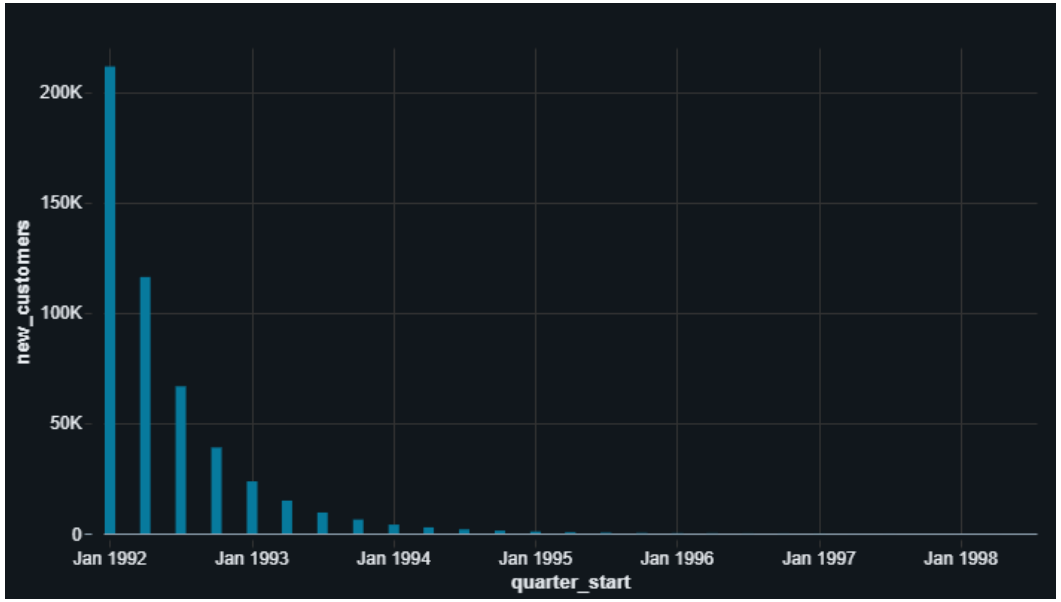


Figure 3: Summary of orders for each quarter

- How the tables connect: customers place orders, orders contain line items, and line items refer to parts and suppliers. The diagram shows which tables can be joined and on which keys, so joins aren't guesswork.
- New customers (quarter of first order): 499,989 in total, 87% of them in 1992.
- Caveat: the data starts in 1992, so early customers all look "new". The fall afterwards is partly an artefact of this

Web Data Challenge: Key Findings (Sessions and their increases in volumes)

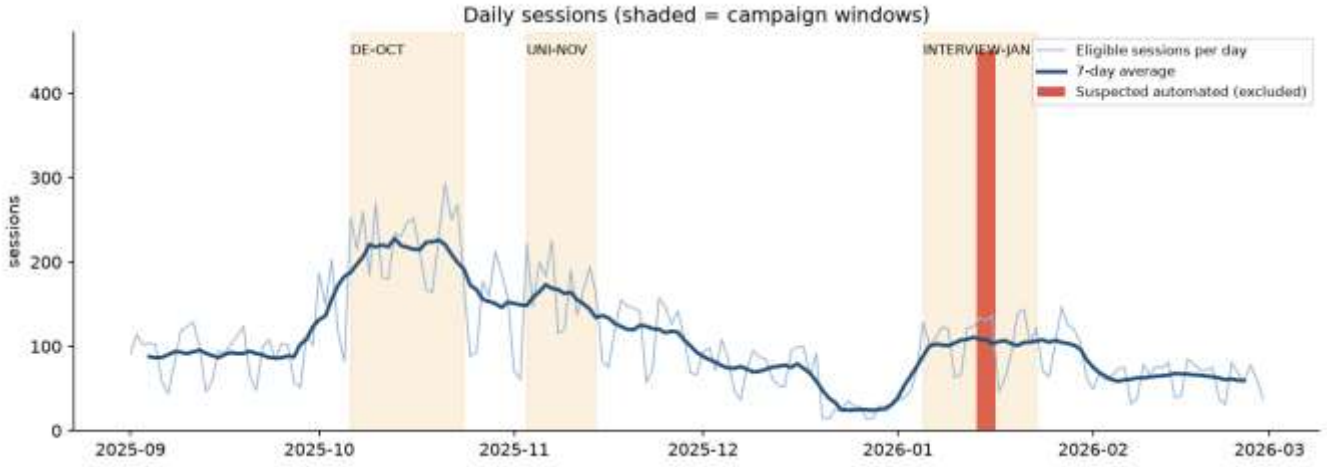


Figure 3: Chart of Daily Sessions from Sep 2025 to Mar 2026

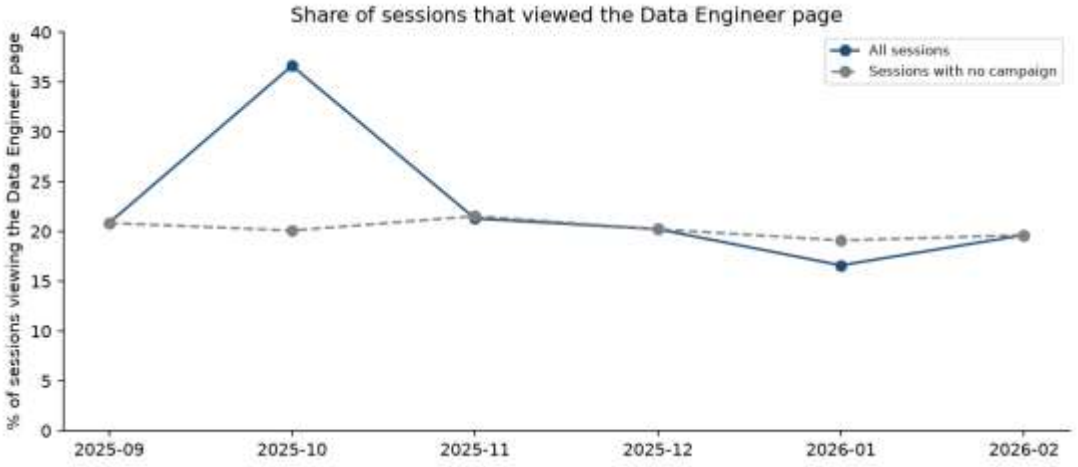


Figure 4: Chart of Interest in Data Engineer role

- October was the peak. Sessions roughly doubled, driven by the LinkedIn Data Engineering campaign (Social visits: 76 a day, up from 19) (Fig.1).
- Email caused smaller rises in November and January. Traffic was lowest from 20 December to 4 January (Fig.1).
- Data Engineer interest jumped to 37% in October, but stayed near 20% in sessions with no campaign (Fig.2).

Web Data Challenge: Key Findings (Unusual periods and segments)

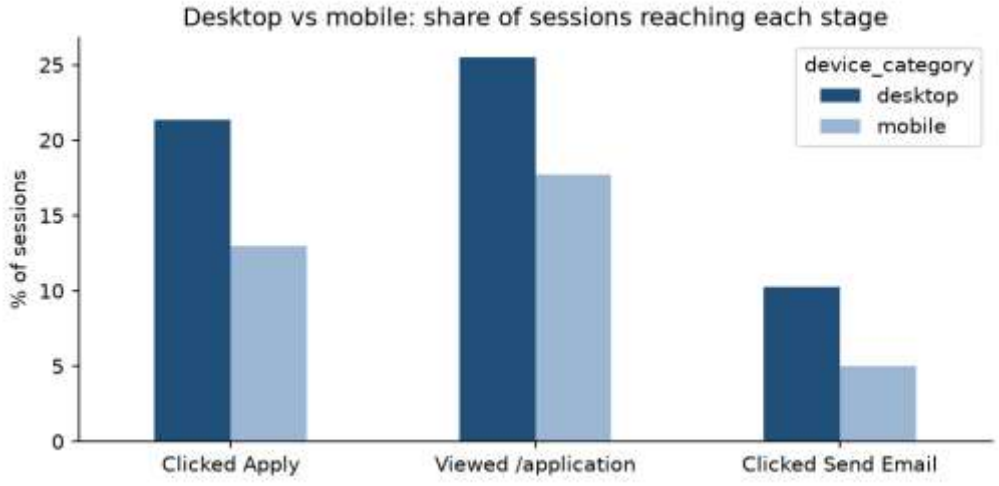


Figure 5: Chart of Desktop Sessions compared to Mobile Sessions

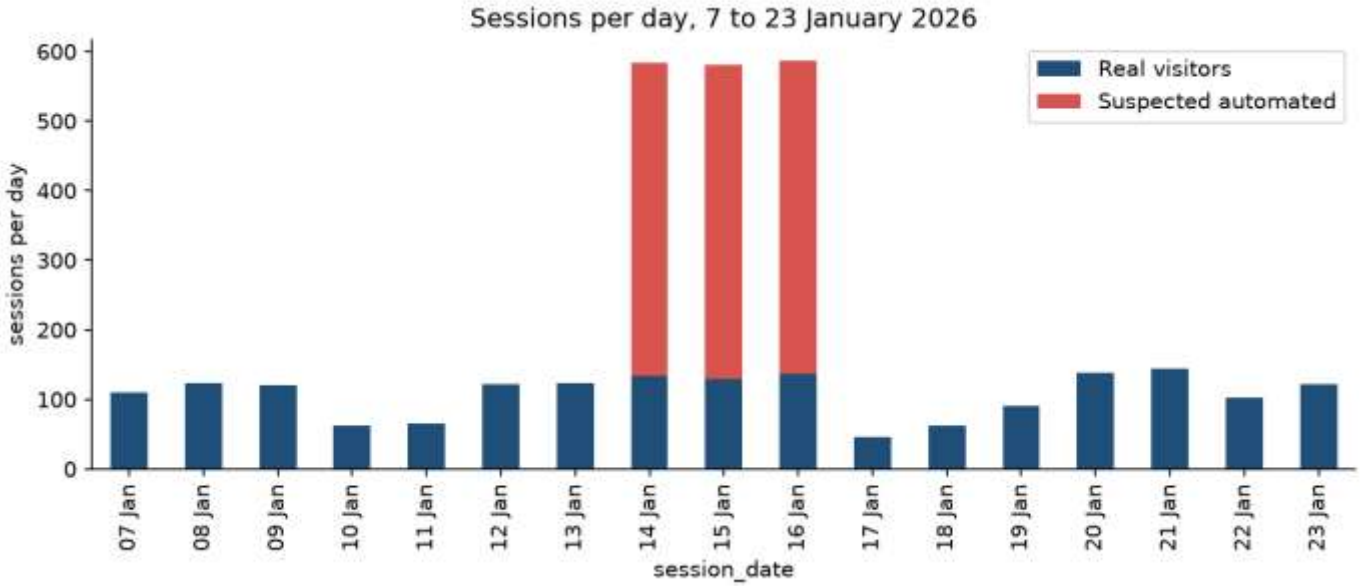


Figure 6: Chart of Unusual traffic volumes in Jan

Unusual periods and segments	Evidence	Possible Reasons
14 to 16 Jan: a burst of sessions (Fig. 3)	450 sessions a day, all desktop, one page view, no clicks	Data issue (Automated Traffic)
6 to 24 October: the Data Engineer campaign boost	Linked to the Data Engineer campaign, but other traffic also rose	Real behaviour linked to campaign and seasonal interest
Desktop versus mobile: a large gap in application intent	Gap at every stage	Preference towards better user experience

Web Data Challenge: Recommendations, Limitations and Use of AI

Recommendations:

- Run role-specific campaigns for the other roles, starting with Technical Analyst.
- Look into the mobile application experience.
- Stop automated traffic from entering the reports and monitor for it.

Limitations and challenges:

During completing this practical task, I faced multiple challenges and limitations:

- The task brief was open-ended, which slowed me down, because I was not sure where to start
- Writing scripts was the biggest challenge for me due to the lack of knowledge
- Choosing the right denominator, in some cases like email click rate and automated sessions, was also challenging
- Data represented narrow selection of users and their sessions (only those who provided consent)

Use of AI:

During completing this practical task, AI was used responsibly to assist me in research and code production processes:

- Generating scripts (Data_Cleaner_SCR, Data_Metrics_SCR; all scripts were reviewed with detailed explanation and commented by myself)
- Researched and produced code for referential integrity
- Generating formulas (Defining an application-intent journey section was suggested to me as support to answering Question 2 and was integrated using AI)